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Introduction

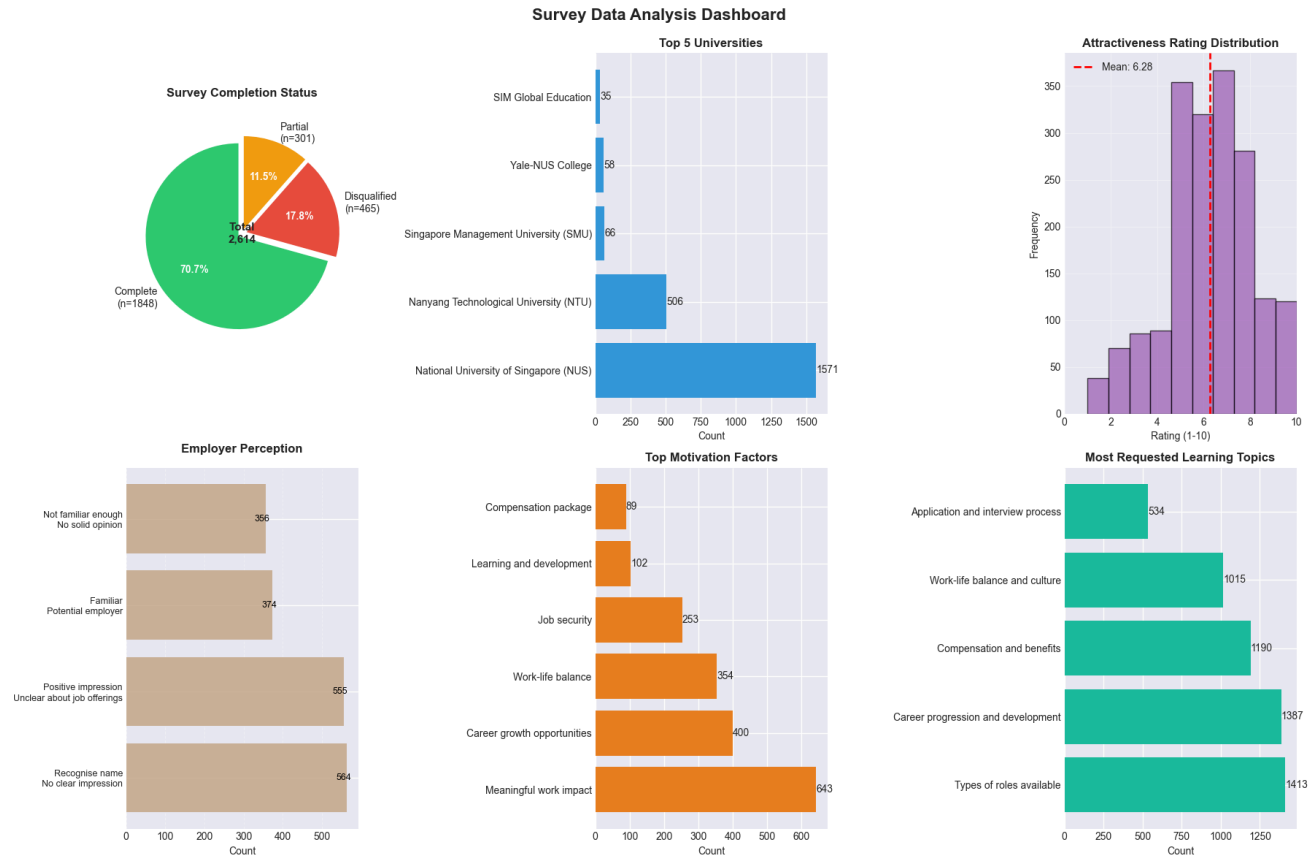
In today's talent-centric landscape, understanding candidate perceptions is paramount, yet deriving clear insights from complex survey data remains a significant challenge for many organisations. DataX's proposed analytics tool transforms this daunting task into a seamless, automated process. **By simply parsing raw survey data into the system, companies unlock a powerful engine that automatically generates comprehensive, actionable intelligence.**

Our solution eliminates the traditional barriers of data science expertise and time-intensive analysis, delivering clear, interpretable insights into survey design flaws, respondent behaviour, and key drivers of employer attractiveness with minimal effort. The following report details how this tool systematically addresses each core business requirement, from identifying question redundancies to segmenting audiences, turning raw responses into a strategic asset for HR and recruitment teams.

Dataset Overview

This dataset contains 2,614 survey responses from university students and graduates, primarily from Singaporean institutions. It includes demographic and academic details such as institution, year of study, field of study, nationality, and gender, as well as perceptions of an employer. Key survey sections cover employer familiarity, areas of interest, attractiveness ratings, and primary motivators for applying. The data contains a mix of complete, partial, and disqualified responses, with some geographic and device metadata included. Overall, it provides insights into student employer preferences and awareness levels.

Phase A: Brief Exploratory Data Analysis (Summary)



Survey Completion & Data Quality

With 1,848 complete responses out of 2,614 total, the survey achieved a 70.7% completion rate. However, the 465 disqualifications (17.8%) and 301 partial responses (11.5%) represent significant attrition, pointing to potential issues with survey length, complexity, or targeting that warrant further investigation.

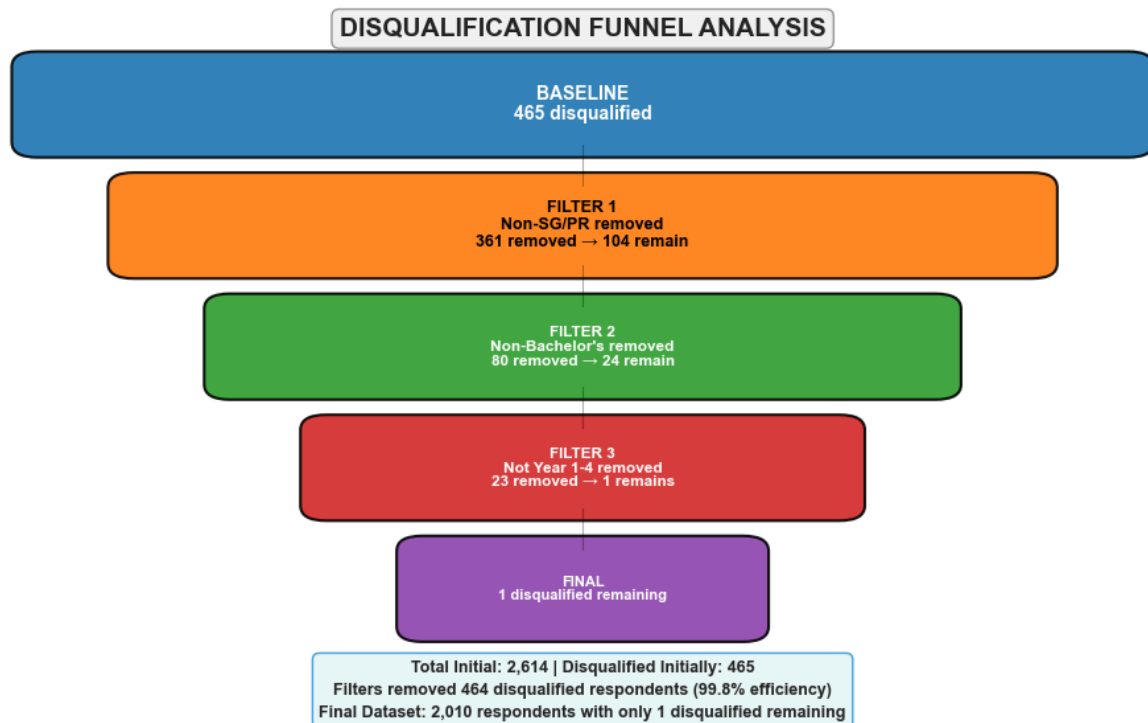
Employer Perception & Attractiveness

The mean attractiveness rating of 6.28 indicates moderate employer appeal. Perception data reveals a critical insight: While brand recognition is high (564 respondents recognise the name), clarity about specific job offerings is low (555 have a positive impression but are unclear about roles). This presents a clear communication gap between general brand awareness and concrete employment understanding.

Candidate Motivations vs. Information Needs

A distinct alignment gap exists between what motivates candidates and what they wish to learn. "Meaningful work impact" is the top motivator (643), yet "Types of roles available" is the most requested learning topic (1413). Similarly, while "Career progression" is a key motivator (400), it is also the second-most requested topic (1387), indicating that current communications are not adequately addressing these fundamental concerns.

Phase B: Disqualification Analysis & Insights



Rationale for Disqualification Analysis

We focus on understanding the 465 disqualified responses to identify survey design weaknesses and respondent behaviour patterns. By examining disqualification triggers, we can pinpoint opportunities to improve respondent targeting and data quality. This analysis will inform both technical refinements to the survey instrument and strategic adjustments to maintain targeted respondent engagement throughout the data collection process.

Results

The disqualification analysis revealed that nearly all disqualified respondents (99.8%) could have been filtered out at the survey's outset using three basic eligibility criteria: nationality (Singaporean/PR), pursuit of a Bachelor's degree, and current year of study (Year 1–4). This finding

indicates that the vast majority of disqualifications stem from mismatched targeting rather than survey content, resulting in significant wasted respondent effort and processing resources.

Recommended Improvements

To improve efficiency and data quality, we recommend immediately implementing upfront qualification questions to auto-disqualify ineligible participants before they proceed.

1. Are you Singaporean or a Permanent Resident?
2. Are you pursuing a Bachelor's degree?
3. Are you in Year 1-4 of your studies?

This change would conserve marketing and operational resources, improve engagement with truly eligible candidates, and lay the groundwork for a medium-term automated pre-screening system. These screening questions can be modified based on the needs of the company and the target group of the respondents.

Guided by these exploratory findings, the subsequent analysis will focus on converting these observations into actionable intelligence for survey designers and HR teams. Having addressed the front-end efficiency of respondent qualification, we now turn our attention to the internal structure of the survey itself. While filtering improves respondent targeting, the survey's effectiveness ultimately depends on its design clarity and question efficacy. The report will now proceed to a detailed investigation across 4 core areas.

1. **Survey Design Diagnostics:** (Methodology)
2. **Factors for Incomplete Surveys:** Centers on Drop-off Analysis and Logistic Regression models to predict the likelihood of abandonment
3. **Respondent Segmentation:** (Methodology)
4. **Drivers of Attractiveness:** (Methodology)

Finally, we will translate all analytical outputs into concrete recommendations for survey redesign and targeted employer branding.

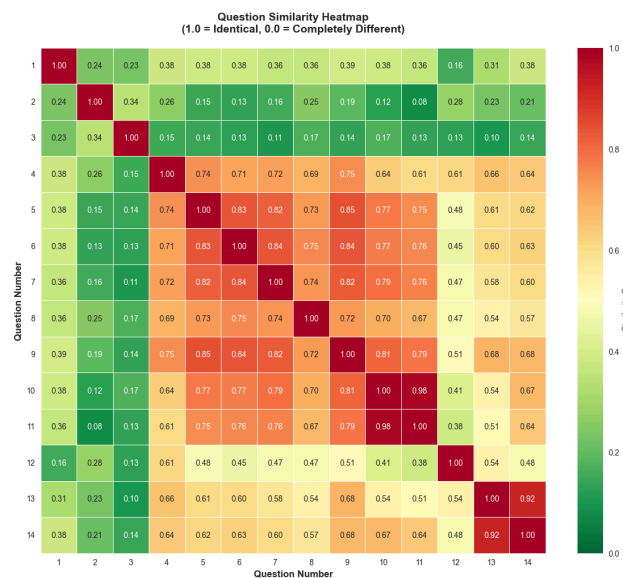
Requirement 1: Identifying patterns, correlations, or redundancies in survey questions that suggest opportunities for improvement

Methodology

This analysis utilised a three-stage pipeline to systematically evaluate and enhance survey question quality.

Stage 1: Semantic Similarity Analysis

- Employed the `all-mpnet-base-v2` sentence transformer model to convert each question into vector embeddings
- Calculated cosine similarity scores (0-1 scale) between all question pairs
- Generated a similarity heatmap to visually identify potential redundancies



Example of Heatmap for Similarity Score

Stage 2: AI-Powered Topic Modelling

- Processed all questions through Google's Gemini Pro model
- Applied natural language understanding to cluster questions into thematic categories
- Generated structured topic groupings based on content and intent

Stage 3: AI Synthesis & Recommendation Generation

- Integrated similarity scores and topic groupings as input to Gemini Pro
- Generated human-readable recommendations covering redundancy elimination, structural improvements, and clarity enhancements
- Applied survey design best practices to create actionable insights

Results & Insights

1. Identified Redundancies & Similar Questions

The semantic analysis revealed several question pairs with high similarity scores (>0.85), indicating potential redundancy:

- Highest Similarity Pair (Score: 0.92): *"What is your current year of study as of 2025?"* and *"What will be your highest qualification when you graduate?"*
Finding: These questions capture overlapping information about academic progression and could be consolidated.
- Moderate Similarity Cluster (Scores: 0.76-0.84): Multiple questions about employer perception and motivation factors showed conceptual overlap, particularly between word association questions and structured perception scales.

2. Thematic Grouping Results

The AI identified 5 logical categories that naturally emerged from the question set:

1. Demographic & Background (5 questions)
 - Institution, year of study, qualification, subject, nationality, gender
2. Employer Perception & Association (4 questions)
 - 3-word associations, perception statements
3. Information Needs & Learning Topics (6 questions)
 - Role types, career progression, compensation, work-life balance, and application process
4. Attractiveness & Motivation (2 questions)
 - Rating scale, motivating factors
5. Session & Technical Data (6 metadata fields)
 - Time, device, location, completion status

3. Recommended Structural Improvements

1. Consolidate Redundant Questions
 - Merge the two academic progression questions into a single multi-part question
 - Review the overlap between word associations and perception scales; consider dropping one format
2. Logical Survey Flow:
 - Opening: Demographic & Background (5 Qs)
 - Employer Perception & Association (4 Qs)

- Information Needs & Learning Topics (6 Qs)
 - Attractiveness & Motivation (2 Qs)
3. Clarity Enhancements:
- Rephrase "Which of these statements best describes your current perception..." to be more specific: "Which statement best matches your view of [Employer] as a place to work?"
 - Add brief context before the 3-word association question: "Think about [Employer]'s reputation, culture, and work environment..."

Finalised Gemini Output (Based on this dataset)

Based on our code and dataset, this is the finalised output by Gemini.

Improvement Insights

1. Duplicate Questions (Score > 0.9)

- **Q1:** Other - Write In (Required):What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Other - Write In (Required):What do you wish to learn more about regarding the organisation as an employer? (Pick 3) - 1
Recommendation: DROP Q2. It appears to be a direct copy of Q1.
- **Q1:** Which of these factors would most motivate you to apply for a position at the organisation? **Q2:** Other - Write In (Required):Which of these factors would most motivate you to apply for a position at the organisation? **Recommendation:** KEEP Q1 and DROP Q2. Q1 likely provides a list of pre-defined factors. Q2 is just a write-in which likely provides the "other" option for Q1, so the write-in is already contained within Q1.

2. Potentially Redundant Questions (0.8 < Score < 0.9)

The following pairs should be reviewed to determine if they are truly measuring different aspects of information interest:

- **Q1:** Types of roles available:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Application and interview process:What do you wish to learn more about regarding the organisation as an employer? (Pick 3)
Suggestion: Keep both if they address distinct points. Consider combining into a single question if the responses are highly correlated and the distinction is not meaningful.
- **Q1:** Career progression and development:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Application and interview process:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Suggestion:** Keep both if the information sought is meaningfully different. A candidate might want to know *specifically* how the application relates to progression.
- **Q1:** Career progression and development:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Compensation and benefits:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Suggestion:** Keep both if they provide distinct insights. If the information desired about career progression is almost always tied to compensation, consider merging.
- **Q1:** Types of roles available:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Career progression and development:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Suggestion:** Ensure the available roles options are distinct from advancement opportunities.
- **Q1:** Compensation and benefits:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Application and interview process:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Suggestion:** They're both likely to be separate, but verify responses.
- **Q1:** Types of roles available:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Compensation and benefits:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Suggestion:** Verify whether the roles offered have an impact on the compensation.
- **Q1:** Application and interview process:What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Q2:** Other - Write In (Required):What do you wish to learn more about regarding the organisation as an employer? (Pick 3) **Suggestion:** See if the "Other" responses can be included within a pre-defined response.

3. Survey Segmentation Recommendation

The survey should be organized into the following sections in this order:

1. **Demographics:** (Nationality, Gender, Subject of Study)
 - Reason: Gathering demographic information first is standard practice. This is non-invasive and sets the stage for more specific questions.
2. **Employer Perception:** (Current Perception, Attractiveness Rating)
 - Reason: Understanding the initial perception allows you to see if the later information interests influence perception.
3. **Information Interest:** (Roles, Career, Compensation, Work Life, Application, Other)
 - Reason: All of these ask about potential candidate desires for information from the employer. Subsections are logical, but could all be one question as well.
4. **Motivation Factors:** (Motivating Factors, Other)
 - Reason: Gathering these last gives the users time to digest all the other questions that preceded this, to properly inform their answer.

4. Question Clarity Improvements

All questions are sufficiently clear.

Requirement 2: Understanding factors associated with partial survey responses

Methodology

The objective of this analysis is to identify and understand the factors associated with partial or incomplete survey responses. By identifying behavioural and environmental triggers for abandonment, companies can refine survey design to maximise completion rates and improve the quality of internal data.

The analysis utilised a dataset containing 2,149 valid responses, as those who were disqualified were not the target respondents and were removed from the analysis. Key variables analysed include:

- **Temporal Factors:** Start time and date, weekend status, and business hour indicators.
- **Technical Metadata:** Device type (Mobile vs. Desktop) extracted from User Agent strings.
- **Demographic Data:** Institution, current year of study, qualification, subject major, nationality, and gender.
- **Survey Progress:** Milestone indicators for sections such as Demographics, Perception, Interests, and Motivation.

The study follows a multi-step analytical framework:

1. **Feature Engineering:** Raw data was transformed into behavioral indicators, such as identifying "zero-duration bounces" and "likely timeouts" (sessions exceeding 15 minutes that remained partial).
2. **Drop-off Analysis:** A survey completion funnel was constructed to identify critical exit points.
3. **Statistical Validation:** Chi-squared tests were applied to categorical features to determine their statistical independence from the completion status.
4. **Interpretability Modelling:** Odd-ratio values were used to determine the impact of situational factors versus demographic noise.

Finally, we used Gemini to summarise the results in an interpretable format.

Results & Insights

Part 2A: Drop-off Analysis (Survey Completion Funnel)

Survey Completion Funnel: Identifying Critical Drop Points



1. Major Initial Drop-off: The most significant loss of respondents occurs between "Started" and "Demographics", with a drop from 2,149 to 1,911 (an 11% loss or 238 respondents). This indicates the opening section, which contains basic filtering or personal questions, is the primary barrier to continuation.
2. Stable Middle Section: After the demographics section, completion rates stabilise remarkably. The progression from Demographics (89%) to Perception (86%) shows only a minimal 3% drop, suggesting that once respondents pass the initial hurdle, they are highly likely to complete the entire survey.
3. High Overall Completion Rate: An 86% completion rate from those who started is very strong for survey data, indicating that the core content is well-designed and engaging for the qualified audience.

We recommend immediately optimising the first 2-3 questions of the survey. Make them simple, clearly relevant, and focused on essential qualifications and demographic information. This change could potentially recover a significant portion of the lost 11% and boost the overall completion rate closer to 95%.

Part 2B: Logistic Regression (Odds Ratio)

Before our logistic regression, we performed statistical validation to remove categorical variables that are insignificant for the completion of survey data. In this case, “is_weekend” and “is_singapore”, as there is no significant difference in abandonment between weekdays and weekends, along with the location of the respondent.

We also performed Outlier Mitigation, removing 232 “Bouncers” who spent less than 10 seconds on the platform. We attribute these to a technical glitch or information overload, consistent with the funnel observed in Part 2A. This refined our dataset to 1,917 high-quality respondents, allowing us to shift the focus from “Technical Noise” to “Active Abandonment” to understand why people who chose to start the survey chose to stop.

The table below illustrates the top 2 factors for survey abandonment, summarised by Gemini.

Factor	Category	Odds Ratio	Interpretation and Risk Level
is_mobile	Technical	1.46	HIGH RISK: Mobile users 46% more likely to partially complete
hour_started	Temporal	1.25	MODERATE RISK: Users who start later are more likely to partially complete

1. Poor mobile optimisation: Phone users being 46% more likely to quit before completion. This indicates the interface is not adapted for smaller screens, causing navigation difficulties and likely frustrating a substantial portion of respondents who access the survey on the go.

Recommendation: Immediate improvements should focus on responsive design: Simplifying layouts, enlarging touch targets, and ensuring faster load times, specifically for mobile devices.

2. Suboptimal timing: Respondents starting the survey later in the day are 25% more likely to partially complete, suggesting issues with fatigue or distraction outside core hours.

Recommendation: Survey invitations can be strategically scheduled during peak engagement windows, ideally between late morning and early afternoon, and the design could be adapted for evening users with shorter formats or clearer time expectations to maintain focus and completion rates.

Finalised Gemini Output (Based on this dataset)

Based on our code and dataset, this is the finalised output by Gemini.

Gemini Insights: Partial Survey Analysis

Okay, based on the logistic regression results provided, here's a summary of the key factors influencing survey abandonment and actionable recommendations to address them:

1. Concise Summary of Top Factors Driving Abandonment:

The primary factors associated with increased survey abandonment are:

- **Mobile Device Usage (is_mobile):** Respondents taking the survey on mobile devices are significantly more likely to abandon it.
- **Start Time (hour_started):** Surveys started at later hours of the day (presumably outside typical business hours) are associated with higher abandonment rates.
- **Gender (gender_fixed_Male):** Male respondents are more likely to abandon the survey.
- **Subject Area (subj_grouped_Business/Management):** Respondents in Business/Management fields are more likely to abandon.

Conversely, the following factors are associated with *lower* abandonment rates:

- Taking the survey during **business hours**
- Being in the **STEM** field.
- Attending a **Non-Autonomous/Other Institution**

2. Actionable Recommendations for the Top 2 Key Factors:

Here are specific, actionable recommendations to address the top two drivers of abandonment (Mobile Device Usage and Start Time):

- **Factor 1: Mobile Device Usage (is_mobile):**
 - **Recommendation: Optimize the survey for mobile devices and test rigorously.** This goes beyond simply being "responsive." It means focusing on usability for smaller screens and touch inputs.
 - **Specific Action:** Implement a mobile-first design approach. This includes:
 - Using larger fonts and buttons.
 - Breaking up long questions into shorter, more manageable chunks.
 - Ensuring adequate spacing between answer options to prevent accidental selections.
 - Thoroughly testing the survey on a variety of mobile devices and operating systems (iOS and Android) to identify and fix any display or functionality issues. This should include testing on older devices as well.
 - Implement a progress bar that clearly shows how far the user is through the survey.
 - Condense as much as possible, avoid unnecessary elements, and make the content easy to read.
 - **Rationale:** Mobile users often have shorter attention spans and are more easily frustrated by poorly designed surveys. A well-optimized mobile experience can significantly reduce abandonment.
- **Factor 2: Start Time (hour_started):**
 - **Recommendation: Send survey invitations and reminders at optimal times, targeting when respondents are most likely to engage.**
 - **Specific Action:** Analyse existing data to determine the hours the survey is started in. If these are outside of business hours, adjust send times for survey emails and text reminders.
 - **Rationale:** People are more likely to complete a survey when they are in a suitable environment and have sufficient time and focus. Sending surveys at suboptimal times (e.g., late at night, early in the morning) increases the likelihood of abandonment.

By focusing on these targeted interventions, you can expect to see improvements in your survey completion rates. Remember to track your completion rates before and after implementing these changes to assess their effectiveness.

Requirement 3: Discovering meaningful demographic or behavioural segments among respondents

Methodology

This analysis employed a structured, three-part methodology to segment respondents based on their perceptions and motivations.

Stage 1: Feature Engineering

- Created composite scores from raw survey responses:
 - Awareness Score (0-1): Normalised from perception statements measuring familiarity with the employer
 - Attractiveness Score (0-1): Normalised 1-10 ratings of employer appeal
 - Composite Intent Score (0-1): Weighted average of Awareness and Attractiveness representing overall conversion potential

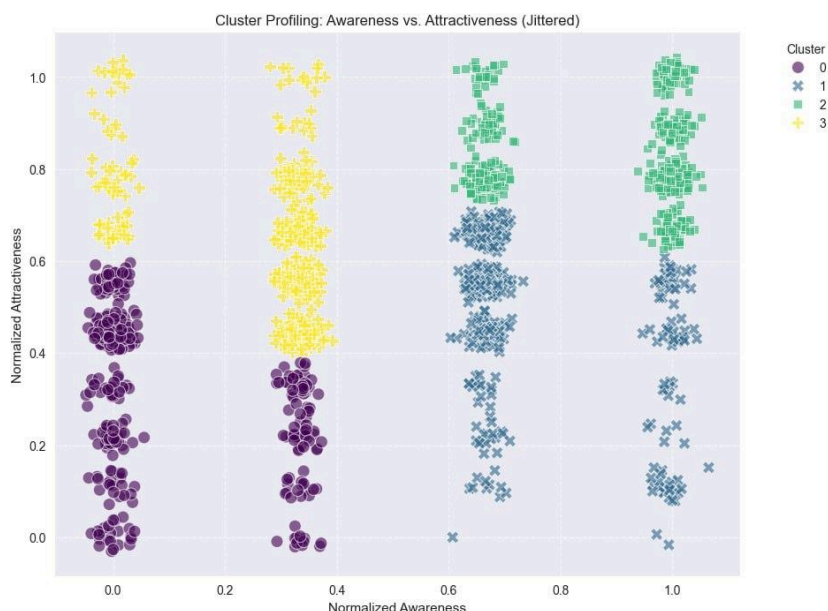
Stage 2: K-Means Clustering (k=4)

- Applied the K-Means algorithm to group respondents based on Awareness and Attractiveness scores
- Selected k=4 clusters based on business logic mapping to recruitment funnel stages
- Visualised results on the Awareness vs. Attractiveness matrix

Stage 3: Cluster-Specific Information Needs Analysis

- Aggregated "Wish to learn more about" responses for each cluster
- Identified top-requested topics per segment
- Mapped findings to targeted communication strategies

Results & Insights



Example of Cluster Profiling

Finalised Gemini Output (Based on this dataset)

Based on our code and dataset, this is the finalised output by Gemini.

Gemini Cluster Analysis

Here's an analysis of the clusters in the scatter plot, along with actionable insights:

1. Cluster Analysis:

- **Cluster 1 (Top Right):** High Awareness, High Attractiveness. These respondents are both well aware of your offering and find it highly appealing.
- **Cluster 2 (Top Left):** Low Awareness, High Attractiveness. These respondents find your offering attractive but have low awareness.
- **Cluster 3 (Bottom Right):** High Awareness, Low Attractiveness. These respondents are aware of your offering, but don't find it attractive.
- **Cluster 4 (Bottom Left):** Low Awareness, Low Attractiveness. These respondents have low awareness and low attractiveness toward your offering.

2. Actionable Insights for Companies:

- **Cluster 1 (High Awareness, High Attractiveness):**
 - **Action:** Focus on customer loyalty programs, advocacy marketing (e.g., referral programs), and upselling/cross-selling. Nurture these respondents and turn them into brand advocates. Gather testimonials and case studies.
 - **Reasoning:** These are already your "fans." Capitalize on their positive sentiment to drive further growth.
- **Cluster 2 (Low Awareness, High Attractiveness):**
 - **Action:** Invest in targeted marketing campaigns focused on increasing awareness. Consider approaches like targeted advertising, social media campaigns, content marketing, or public relations.
 - **Reasoning:** These respondents are potential customers who simply need to know more about what you offer. Highlight the features they will find attractive, and make the offering more visible.
- **Cluster 3 (High Awareness, Low Attractiveness):**
 - **Action:** Conduct a thorough review of your offering. Gather feedback on why these respondents find it unattractive. This might involve product improvements, pricing adjustments, or better communication of value. Consider repositioning the offering in the marketplace.
 - **Reasoning:** Something is preventing these aware respondents from converting. Address the issues that make it unattractive.
- **Cluster 4 (Low Awareness, Low Attractiveness):**
 - **Action:** This group is the least promising in the short term. Consider whether it's worth investing significant resources into this segment. Potentially, focus on broader brand-building efforts that may indirectly improve awareness and attractiveness over time. Alternatively, revisit your target audience definition.
 - **Reasoning:** Resources are generally better spent on segments with higher potential for conversion.

4. Key Information Needs by Cluster:

- **Cluster 1:** Focus on reinforcing positive perceptions and providing excellent ongoing service. Companies should prioritize content that reinforces the value proposition, showcasing success stories, and providing exclusive benefits.
- **Cluster 2:** Prioritize content that builds awareness and clearly communicates the value proposition. Information should be easily accessible and engaging, highlighting the most attractive features and benefits.
- **Cluster 3:** Focus on understanding and addressing the reasons for the lack of attractiveness. This involves gathering feedback, conducting market research, and transparently communicating any changes made to address concerns.
- **Cluster 4:** Focus on generating overall brand awareness and differentiating your offering from competitors. Information should be broad, informative, and aim to create a positive first impression. Longer-term brand building is key here.

Saved clustered data to: clustered_respondents.csv

```
--- Running Information Needs Analysis ---
Top 3 Information Needs:
1. Roles: 1413 respondents
2. Career: 1387 respondents
3. Compensation: 1190 respondents
```

Requirement 4: Extract insights from the applicant's perspective on what drives employer attractiveness

Methodology

The feature to meet this requirement is an end-to-end insights generation pipeline to understand *what drives employer attractiveness from the applicant's perspective*. The objective is to systematically surface interpretable, evidence-based insights that can inform employer branding and talent strategy.

The methodology consists of three main stages:

1. Data preparation and feature construction
2. Insight discovery
3. Automated insight narration using a large language model (LLM)

1. Data Inputs and Feature Construction

The survey data includes an **employer attractiveness rating (1–10)**, multi-select responses on **motivation to apply** and **topics respondents wish to learn more about**, a **categorical perception signal** describing familiarity with the organisation, and **demographic attributes** such as discipline and institution type.

All categorical and multi-select fields are transformed into **binary indicators** for interpretability:

- **motivate_*** variables capture reasons for applying,
- **learn_*** variables capture information gaps,
- **perception_*** variables represent distinct familiarity states (one-hot encoded, non-ordinal),
- **ctrl_*** variables represent demographic characteristics as binary flags.

This encoding enables transparent, one-dimensional comparisons (group vs rest) without complex interactions.

2. Insight Discovery Framework

The analysis is structured into five parts, each addressing a specific aspect of employer attractiveness:

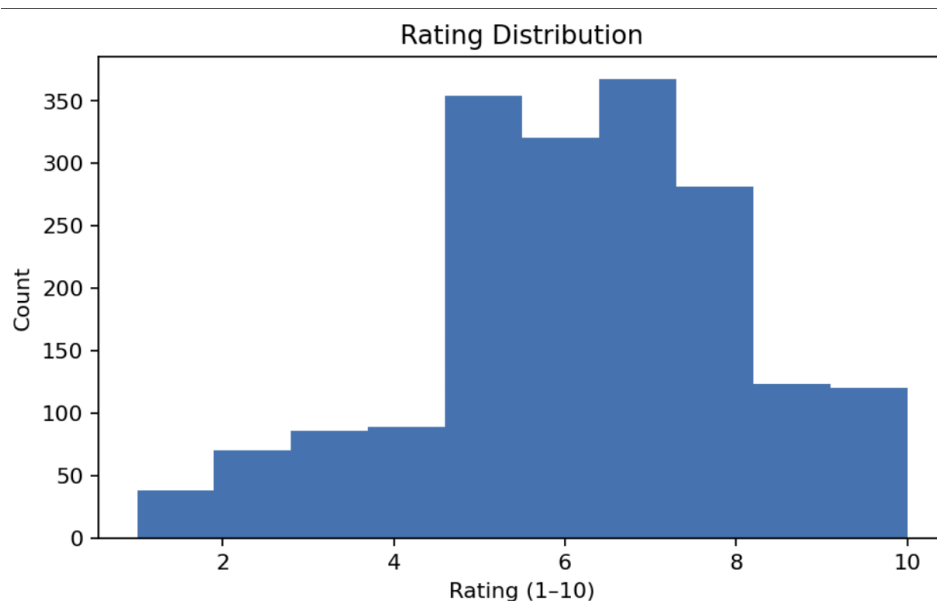
- **Part A: Overall Attractiveness Landscape**
Establishes a baseline by summarising the distribution of employer attractiveness ratings, providing context for all subsequent analyses.
- **Part B: Motivation Drivers**
Identifies the most common motivations for applying and highlights which motivations are associated with more positive employer perceptions.
- **Part C: Learning & Information Gaps**
Examines what applicants want to learn more about, how learning needs vary by perception state, and which topics exhibit higher uncertainty rather than dissatisfaction.
- **Part D: Demographic Differences in Motivation and Learning**
Uses binary group-versus-rest comparisons to assess whether motivations and learning needs differ meaningfully across demographics, reporting only the strongest contrasts or explicitly noting when none exist.
- **Part E: Demographic Differences in Ratings**
Compares average employer attractiveness ratings across demographic groups to identify perceptual gaps, with null findings reported explicitly.

3. Insight Packaging and Automation

- Outputs from Parts A–E are compiled into a **machine-generated Markdown insight pack** containing summary bullets, tables, and chart references. While analytically complete, this file is not directly user-friendly.
- A **custom prompt file** is then generated to guide a large language model (LLM), explaining the purpose of each part, when sections may be omitted, and how to rewrite the content clearly without inventing insights.
- Finally, an **LLM API call (Gemini)** ingests the prompt and insight pack to translate technical outputs into a **clear, concise Markdown report**, explaining charts, surfacing key takeaways, and improving readability for non-technical audiences.

Results

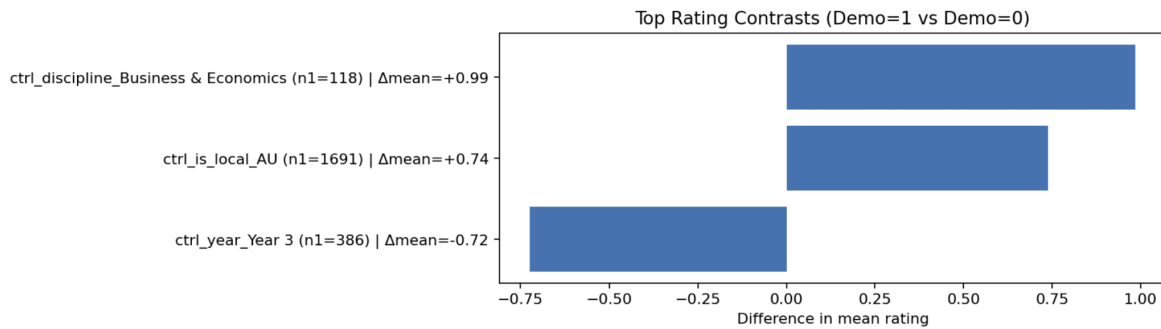
- Overall attractiveness
 - Employer attractiveness is moderately positive, with ratings centred around 6.3 / 10.
 - Ratings show moderate dispersion, indicating generally favourable but varied perceptions.



- Motivation drivers
 - Applicants are primarily motivated by meaningful work impact.
 - Some motivations are associated with higher employer attractiveness ratings, acting as differentiators.
- Learning & information gaps
 - The strongest information gaps relate to career development and growth opportunities.
 - Less familiar respondents exhibit broader learning needs, suggesting uncertainty rather than dissatisfaction.
 - Clearer communication on career pathways could improve perceptions.
- Demographic patterns in motivation & learning
 - No meaningful demographic differences were found.
 - Motivation and learning needs are broadly consistent across groups, supporting the generalisability of insights.

- Demographic patterns in ratings

- Business & Economics students rate the organisation ~1.0 point higher on average.
- Local autonomous university respondents rate the organisation ~0.7 points higher.
- Year 3 students rate the organisation ~0.7 points lower than others.



- Overall takeaway

- Employer attractiveness is driven by shared motivations and common information gaps.
- However, perceptions vary across specific applicant segments, indicating scope for targeted employer branding and communication.

Insights

These gaps suggest perceptual differences that may reflect varying expectations or exposure, and they highlight specific segments where employer messaging may need adjustment.

The analysis shows that employer attractiveness is driven more by shared motivations and information needs than by demographic differences, but perceptions of attractiveness itself do vary across certain groups.

Taken together, the results suggest that:

- Strengthening communication around career development can improve attractiveness broadly.
- Targeted messaging may be required for specific groups that currently rate the organisation less favourably.

Finalised Gemini Output (Based on this dataset)

Based on our code and dataset, this is the snippet of the finalised output by Gemini.

```
# Employer Attractiveness Report

**Executive Summary:**
- Overall attractiveness ratings are moderately positive, with a mean of 6.28 (on a scale of 1-10).
- The most popular motivations for applying are "career growth", "company culture", and "purpose/impact".
- Selecting motivations such as "innovative culture" and "high-performance environment" is associated with higher attractiveness ratings.
- The most common learning needs are around "career paths" and "day-to-day work".
- There is a need to address information gaps around "promotion paths", as needing to know more is associated with greater rating variability (uncertainty).
- Respondents in Business & Economics rate the organisation more favourably, whereas Year 3 respondents rate it lower.

## Part A — Rating Landscape (Overall Attractiveness)
- *What this section contains: The distribution of employer attractiveness ratings.
- *Key takeaways: Ratings are moderately positive, but with noticeable spread.
  - The mean rating is 6.28, with a median of 6.00 (on a 1-10 scale).
  - The interquartile range (IQR) is 3.00, indicating moderate dispersion.
- *Evidence: See the distribution in Figure A_rating_hist.png.

## Part B — Motivation Drivers (Why applicants apply)
- *What this section contains: The popularity of different motivations for applying and their relationship to attractiveness ratings.
- *Key takeaways: Understanding and catering to top motivations can boost attractiveness.
  - The top 3 motivations account for 75.6% of all selections.
  - "Career growth" is the most popular motivation (34.8% selection rate).
  - Selecting "innovative culture" and "high-performance environment" are associated with higher ratings.
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Conclusion

Our analysis demonstrates the transformative potential of an automated insight generation tool for survey data. Our key value proposition is profound simplicity: **organisations only need to supply their data**. The system then autonomously executes the entire analytical workflow—diagnosing survey design inefficiencies, pinpointing dropout causes, segmenting respondent pools, and decoding the core drivers of employer appeal, before synthesising everything into a clear, stakeholder-ready narrative.

This end-to-end automation simplifies data analysis, making advanced analytics accessible and immediately actionable. By converting raw datasets into prioritised recommendations for survey optimisation and targeted employer branding, this tool empowers companies to make faster, more informed decisions, ultimately enhancing their recruitment effectiveness and strengthening their employer value proposition with unprecedented ease.

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